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Inventor(s)

CARLO, III; Robert Michael et al.

TARGETING GUIDE

Abstract

Disclosed are bone screw drill targeting guides and methods for using such targeting guides that are useful in surgical procedures for correcting hallux valgus deformity.

Inventors: CARLO, III; Robert Michael (Lakeland, TN), AWTREY; George Matthew (Bartlett, TN)

Applicant: WRIGHT MEDICAL TECHNOLOGY, INC. (Memphis, TN)

Family ID: 85176879

Assignee: WRIGHT MEDICAL TECHNOLOGY, INC. (Memphis, TN)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation filed under 37 C.F.R. § 1.53 claiming the benefit under 35 U.S.C. § 120 of any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application, including U.S. patent application Ser. No. 17/814,606, filed Jul. 25, 2022, and U.S. Provisional Patent Application No. 63/231,449, filed Aug. 10, 2021, and are hereby incorporated by reference in accordance with 37 C.F.R. §§ 1.57; 1.97; and 1.98 in their entireties.

FIELD OF DISCLOSURE

[0002] The present disclosure relates to an instrument for guiding a screw or a wire through bone or bones toward a desired target point that can be used in surgical procedures. for correcting hallux valgus deformity.

BACKGROUND

[0003] Guiding a screw or a wire through a bone or multiple bones and/or joints is difficult especially for less experienced surgeons because in many procedures, the target point for the screw or wire is inside a bone and thus not visible. For example, during a minimally invasive surgical procedures such as Minimally Invasive Chevron and Akin osteotomy (MICA) procedure for correcting hallux valgus deformity, a Chevron osteotomy is made in the first metatarsal bone separating the head portion of the first metatarsal from the remainder of the metatarsal. The metatarsal head is then shifted laterally and fixed with two screws. K-wires are traditionally used to hold the metatarsal head at the intended translated position during the subsequent screw fixation procedure. Achieving the desired k-wire trajectory can be difficult. Therefore, a guiding instrument for setting the trajectory of the K-wire in such instance is desired.

SUMMARY

[0004] Disclosed herein is a targeting guide instrument that is configured to allow a surgeon to place a guide wire or a screw at any orientation and it will always pass through a predetermined location in relation to the guide instrument. Thus, by matching the predetermined location of the guide instrument to be the desired target point inside a bone, for example, the surgeon can insert a guide wire or a screw to that target point guided by the guide instrument. The disclosed targeting guide can be useful for placing a guide wire, such as a K-wire, or a screw through two adjacent bone pieces. The two adjacent bone pieces can be two portions of a bone or two adjacent bones.

[0005] The targeting guide comprises: a first guide sleeve defining a first longitudinal axis; a second guide sleeve defining a second longitudinal axis; an articulating arm connecting the first guide sleeve and the second guide sleeve; where the first longitudinal axis and the second longitudinal axis intersect at an intersection point in space and form an intersecting angle, the intersection point being located at a first distance from the first guide sleeve and at a second distance from the second guide sleeve; where manipulating the articulating arm changes the intersecting angle while maintaining the same intersection point; and where the articulating arm is configured to be able to change the first distance.

[0006] A surgical instrument kit is also disclosed, where the kit comprises at least one guide wire and a targeting guide of the present disclosure. A surgical instrument kit comprising at least one guide wire, at least one drill bit, and a targeting guide of the present disclosure is also disclosed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] The inventive concepts of the present disclosure will be described in more detail in conjunction with the following drawing figures. The structures in the drawing figures are illustrated

schematically and are not intended to show actual dimensions.

[0008] FIG. **1** is an isometric view of the targeting guide of the present disclosure.

[0009] FIG. **2** is plan view of the targeting guide of FIG. **1**.

[0010] FIGS. **3-5** are side views of the targeting guide of the present disclosure.

[0011] FIG. **6** is an illustration showing the targeting guide of the present disclosure where the first guide sleeve has been slipped over the first guide wire **W1**.

[0012] FIG. **7** is an illustration showing the targeting guide of the present disclosure where the second guide wire **W2** has been shot through the first and second bone pieces **B1**, **B2**.

DETAILED DESCRIPTION

[0013] This description of the exemplary embodiments is intended to be read in connection with the accompanying drawings, which are to be considered part of the entire written description. The drawing figures are not necessarily to scale and certain features may be shown exaggerated in scale or in somewhat schematic form in the interest of clarity and conciseness. In the description, relative terms such as “horizontal,” “vertical,” “up,” “down,” “top” and “bottom” as well as derivatives thereof (e.g., “horizontally,” “downwardly,” “upwardly,” etc.) should be construed to refer to the orientation as then described or as shown in the drawing figure under discussion. These relative terms are for convenience of description and normally are not intended to require a particular orientation. Terms including “inwardly” versus “outwardly,” “longitudinal” versus “lateral” and the like are to be interpreted relative to one another or relative to an axis of elongation, or an axis or center of rotation, as appropriate. Terms concerning attachments, coupling and the like, such as “connected” and “interconnected,” refer to a relationship wherein structures are secured or attached to one another either directly or indirectly through intervening structures, as well as both movable or rigid attachments or relationships, unless expressly described otherwise. When only a single machine is illustrated, the term “machine” shall also be taken to include any collection of machines that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. The term “operatively connected” is such an attachment, coupling or connection that allows the pertinent structures to operate as intended by virtue of that relationship. In the claims, means-plus-function clauses, if used, are intended to cover the structures described, suggested, or rendered obvious by the written description or drawings for performing the recited function, including not only structural equivalents but also equivalent structures.

[0014] Referring to FIGS. **1** through **5**, a targeting guide **100** is disclosed that is configured to allow a surgeon to place a guide wire **W2** or a screw at any orientation that will always pass through a predetermined location, a target location, in relation to the targeting guide **100**. The targeting guide **100** comprises a first guide sleeve **101** defining a first longitudinal axis **Axis1**, a second guide sleeve **102** defining a second longitudinal axis **Axis2**, and an articulating arm connecting the first guide sleeve **101** and the second guide sleeve **102**.

[0015] The first longitudinal axis **Axis1** and the second longitudinal axis **Axis2** intersect at an intersection point **T**, that is the predetermined location in space, so as to form an intersecting angle β . The intersection point **T** is located at a first distance **D1** (labeled in FIG. **2**) from the first guide sleeve **101** and at a second distance **D2** (labeled in FIG. **2**) from the second guide sleeve **102**.

Flexing the articulating arm changes the intersecting angle β while maintaining the intersection point **T** at the first distance **D1** from the first guide sleeve **101** and at the second distance **D2** from the second guide sleeve **102**. In other words, the intersection point **T** remains at the same point in space relative to the first guide sleeve **101** and the second guide sleeve **102** while the articulating arm is flexed to change the intersecting angle β . This is graphically explained in FIG. **5**. The dotted lines **Axis2a**, **Axis2b**, **Axis2c**, and **Axis2d** illustrate some examples of different orientations that the second longitudinal axis **Axis2** can be placed by flexing and manipulating the articulating arm. As can be seen, no matter what orientation the second longitudinal axis **Axis2** is in, it always intersects the first longitudinal axis **Axis1** at the intersection point **T**.

[0016] Referring to FIG. 7, this feature of the targeting guide **100** is useful in situations where a guide wire **W2** needs to be placed through a first bone piece **B1** and into a second bone piece **B2** toward a predetermined target point **TP** where the target point **TP** is inside the second bone piece **B2** and is not visible to the surgeon. Because the second guide sleeve **102** always points toward the intersection point **T**, which is located along the first longitudinal axis **Axis1** at the first distance **D1** from the tip of the first guide sleeve **101**, by placing the first guide sleeve **101** against the second bone piece **B2** so that the first longitudinal axis **Axis1** goes through the predetermined target point **TP** and that the tip of the first guide sleeve **101** is at the first distance **D1** from the target point **TP**, the intersection point **T** of the targeting guide **100** and the target point **TP** in the second bone **B2** will be the same point in space. Then the guide wire **W2** extending through the second guide sleeve **102** will always be aimed at the predetermined target point **TP** no matter the angular orientation the second guide sleeve **102**. This ensures that the surgeon will always hit the predetermined target point **TP** with the guide wire **W2**.

[0017] Additionally, the articulating arm is configured to be able to change the first distance **D1** to **D1'** as shown in FIG. 6. This adjustability allows one to change the location of the intersection point **T** with respect to the first guide sleeve.

[0018] In some embodiments of the targeting guide **100**, the articulating arm comprises a first pair of linkages **110**, **111** that are arranged parallel to each other and a second pair of linkages **120**, **121** that are also arranged parallel to each other. Each of the linkages **110**, **111** in the first pair have a first end and a second end, the first ends being hingeably connected to the first guide sleeve **101** at respective first hinge points **P1**, **P3**. Each of the linkages **120**, **121** in the second pair have a first end and a second end, the first end being hingeably connected to the second guide sleeve **102** at respective second hinge points **P6**, **P7**. The second ends of the linkages **110**, **111** in the first pair are arranged and hingeably connected to the second ends of the linkages **120**, **121** in the second pair to form a pair of nested hinged joints that allow the two pairs of linkages to flex and form a flex angle θ of less than 180 degrees while keeping the two linkages in each pair to remain parallel to each other. The first of the two nested hinged joints are formed by one linkage **110**, **120** from each pair being connected. The second of the two nested hinged joints are formed by another linkage **111**, **121** from each pair being connected.

[0019] Referring to FIGS. 3-4, the first of the pair of nested hinged joints formed by the linkages **110** and **120** is referred to as an outer joint and the second of the pair of hinged joints formed by the two linkages **111** and **121** is referred to as an inner joint. The hinged joint formed by the linkages **111** and **121** is referred to as an inner joint because that hinged joint is nested inside the hinged joint formed by the two linkages **110** and **120**. The hinged joint formed by the linkages **110** and **120** is referred to as an outer joint because that hinged joint is outside the hinged joint formed by the two linkages **111** and **121**. Because the first pair of linkages **110** and **111** are parallel and the second pair of linkages **120** and **121** are parallel, the two nested hinged joints form the same flex angle θ .

[0020] In some embodiments of the targeting guide **100**, the linkage **120** forming the outer joint is configured so that the length **L** (labeled in FIG. 4) between the nested hinge joint **P5** and the second hinge point **P7** can be adjusted to change the first distance **D1** to **D1'** (FIG. 6). The linkage **120** can be provided with a slot **122** in which the nested hinge joint **P5** can slide to increase or decrease the length **L**. The sliding arrangement between the nested hinge joint **P5** and the linkage **120** can be configured so that a thumbwheel **160** can be used to make the nested hinge joint **P5** slide within the slot **122** and adjust the length **L**. For example, making the length **L** shorter will cause the second guide sleeve holder **102a** to pivot about the hinge joints **P6** and **P7** and change the attitude of the second guide sleeve **102** so that the second longitudinal axis **Axis2** defined by the second guide sleeve **102** to tilt in the direction of the arrow **B** shown in FIG. 4. This will move the intersection point **T** to move away from the first guide sleeve **101** thus increasing the first distance **D1** to **D1'** and decrease the intersecting angle β . On the other hand, making the length **L** longer will cause the second guide sleeve holder **102a** to pivot about the hinge joints **P6** and **P7** in the opposite direction

and change the attitude of the second guide sleeve **102** so that the second longitudinal axis Axis2 defined by the second guide sleeve **102** to tilt in the direction of the arrow A shown in FIG. 4. This will move the intersection point T to move toward the first guide sleeve **101**, thus, decreasing the first distance D1 and increase the intersecting angle β . This feature can be used to adjust the location of the predetermined target point TP inside the target bone piece.

[0021] Referring to FIG. 1, in some embodiments of the targeting guide **100**, the first guide sleeve **101** is configured to receive a surgical guide wire W1. In some embodiments, the second guide sleeve **102** is configured to receive a surgical guide wire.

[0022] In some embodiments of the targeting guide **100**, the first guide sleeve **101** is configured to receive and guide a drill bit. In some embodiments, the second guide sleeve **102** is configured to receive and guide a drill bit.

[0023] The associated surgical method of using the targeting guide **100** will now be described. First, a guide wire W1, such as a K-wire, is placed into the bone piece B2 at a location so that the guide wire W1 is either pointing at the predetermined target point TP inside the second bone piece B2 or the guide wire W1 is extending through the predetermined target point TP. In other words the predetermined target point TP is axially aligned with the guide wire W1 so that the predetermined target point TP is somewhere along the longitudinal axis of the guide wire W1.

[0024] Then, the first guide sleeve **101** is placed over the guide wire W1 and positioned along the guide wire W1 so that the intersection point T of the targeting guide **100** coincides with the predetermined target point TP. As described above the intersection point T of the targeting guide **100** is located along the first longitudinal axis Axis1 of the first guide sleeve **101** at the first distance D1 from the tip of the first guide sleeve **101**. At this point the arrangement looks something like shown in FIG. 6.

[0025] Next, the targeting guide **100** is adjusted so that the second guide sleeve **102** is in a position with an orientation that represents the orientation of the bone screw that the surgeon would like to insert through the first bone piece B1 and into the second bone piece B2. Once the second guide sleeve **102** is in position, a second guide wire W2 can be shot through the second guide sleeve **102** toward the intersection point T which will be at the predetermined target point TP.

[0026] Once the guide wire W2 is placed in position, the targeting guide **100** and the first guide wire W1 can be removed and a cannulated drill bit can be slipped over the guide wire W2 and used to drill a hole into the bone pieces B1, B2. Alternatively, a cannulated bone screw can be slipped over the guide wire W2 and screwed into the bone pieces B1, B2.

[0027] In some other embodiments, after the second guide sleeve **102** is in position, rather than shooting the second guide wire W2 through the second guide sleeve **102**, a pilot hole for a screw can be drilled through the bone pieces B1, B2 using the second guide sleeve **102** as a drill guide. A bone screw can then be threaded into the pilot hole to secure the two bone pieces B1, B2 together. Alternatively, without drilling a pilot hole, a self-tapping bone screw can be inserted through the second guide sleeve **102** and driven into the bone pieces B1, B2.

[0028] The targeting guide **100** and the associated method can be applied to such procedure as Minimally Invasive Chevron and Akin osteotomy (MICA) procedure for correcting hallux valgus deformity where the head portion and the reminder portion of the first metatarsal bone after the Chevron osteotomy are the second bone piece B2 and the first bone piece B1, respectively. After the metatarsal head is shifted laterally, the targeting guide **100** of the present disclosure can be used to fix the two bone pieces using two screws.

[0029] According to an aspect of the present disclosure, a surgical instrument kit is also disclosed, where the kit comprises at least one guide wire and a targeting guide of the present disclosure. A surgical instrument kit comprising at least one guide wire, at least one drill bit, and a targeting guide of the present disclosure is also disclosed.

[0030] Although the devices, kits, systems, and methods have been described in terms of exemplary embodiments, they are not limited thereto. Rather, the appended claims should be

construed broadly, to include other variants and embodiments of the devices, kits, systems, and methods, which may be made by those skilled in the art without departing from the scope and range of equivalents of the devices, kits, systems, and methods.

Claims

1. A targeting guide comprising: a first guide sleeve defining a first axis; a second guide sleeve defining a second axis; an articulating arm including a first pair of parallel linkages and a second pair of parallel linkages arranged with one another so as to connect the first guide sleeve and the second guide sleeve; wherein the first axis and the second axis define an intersection point in space so as to form an intersecting angle, the intersection point being located at a first distance from the first guide sleeve and at a second distance from the second guide sleeve; wherein actuating at least one of the first pair of parallel linkages and the second pair of parallel linkages changes the intersecting angle while maintaining the same intersection point.
2. The targeting guide of claim 1, wherein the first pair of parallel linkages have a first end and a second end, the first end being hingeably connected to the first guide sleeve at a first hinge point, wherein each of the second pair of parallel linkages has first end and a second end, the first end being hingeably connected to the second guide sleeve at a second hinge point, and further wherein the second ends of the linkages in the first pair of parallel linkages are arranged and hingeably connected to the second ends of the linkages in the second pair of parallel linkages to form a pair of nested hinged joints that allow the first and second pairs of linkages to flex and form a flex angle of less than 180 degrees while keeping the first pair of parallel linkages and the second pair of parallel linkages parallel to each other.
3. The targeting guide of claim 2, wherein a first portion of each of first and second pairs of linkages are connected to form a first of the pair of nested hinged joints and a second portion of the first and second pairs of linkages are connected to form a second of the pair of nested hinged joints so that the first of the pair of nested hinged joints is an inner joint and the second of the pair of nested hinged joints is an outer joint.
4. The targeting guide of claim 1 wherein the articulating arm is configured to be able to change the first distance.
5. The targeting guide of claim 2, wherein one of the first pair of parallel linkages and the second pair of parallel linkages in the second pair forming the outer joint is configured so that its length between the nested hinged joint and the second hinge point can be adjusted to change the first distance.
6. The targeting guide of claim 1, wherein the first guide sleeve is configured to receive a surgical guide wire.
7. The targeting guide of claim 1, wherein the first guide sleeve is configured to receive a surgical guide wire.
8. The targeting guide of claim 1, wherein the second guide sleeve is configured to receive and guide a drill bit.
9. The targeting guide of claim 1, wherein the second guide sleeve is configured to receive and guide a drill bit.
10. A targeting guide comprising: a first sleeve defining a first axis and configured to receive a surgical guide wire; a second sleeve defining a second axis and configured to receive and guide a drill bit; an articulating arm including a first pair of parallel links and a second pair of parallel links arranged so as to operatively connect the first sleeve and the second sleeve; wherein the first axis and the second axis intersect at a predetermined point to form an intersecting angle therebetween located at a first distance from the first sleeve and at a second distance from the second sleeve; wherein actuating the first pair of parallel links and the second pair of parallel links changes the intersecting angle and the first distance while maintaining the same intersection point.

- 11.** The targeting guide of claim 10, wherein each of the first pair of parallel links having a first end and a second end, the first end being hingeably connected to the first sleeve at a first hinge point and each of the second pair of parallel links having a first end and a second end, the first end being hingeably connected to the second sleeve at a second hinge point.
 - 12.** The targeting guide of claim 11, wherein the second ends of the first pair of parallel links are arranged and hingeably connected to the second ends of the second pair of parallel links so as to form a pair of nested hinged joints that allow the two pairs of linkages to flex and form a flex angle of less than 180 degrees while keeping the two linkages in each pair to remain parallel to each other.
 - 13.** The targeting guide of claim 12, wherein one linkage from each pair of parallel links is connected to form a first of the pair of nested hinged joints and another linkage each pair of parallel links are connected to form a second of the pair of nested hinged joints.
 - 14.** The targeting guide of claim 12, wherein the first of the pair of nested hinged joints is an inner joint and the second of the pair of nested hinged joints is an outer joint.
 - 15.** The targeting guide of claim 10, wherein one of the links in the second pair of parallel links forming the outer joint being configured so that its length between the nested hinged joint, it forms, and the second hinge point can be adjusted to change the first distance.
 - 16.** A targeting guide comprising: a first sleeve defining a first axis; a second sleeve defining a second axis; an articulating arm including a first pair of parallel links having a first end and a second end, the first end being hingeably connected to the first sleeve at a first hinge and a second pair of parallel links having a first end and a second end, the first end being hingeably connected to the second sleeve at a second hinge such that the first and second linkages are arranged so as to operatively connect the first sleeve and the second sleeve; wherein the first axis and the second axis intersect at a predetermined intersection point so as to form an intersecting angle therebetween, the intersection point being located at a first distance from the first sleeve and at a second distance from the second sleeve; wherein actuating the first pair of parallel links and the second pair of parallel links changes the intersecting angle and the first distance while maintaining the same intersection point.
 - 17.** The targeting guide of claim 16, wherein the second ends of the links in the first pair are arranged and hingeably connected to the second ends of the links in the second pair to form a pair of nested hinged joints that allow the two pairs of links to flex so as to form a flex angle of less than 180 degrees while maintaining the links in each pair parallel to each other.
 - 18.** The targeting guide of claim 17, wherein one link from each pair of parallel links is connected to form a first of the pair of nested hinged joints and another link from each pair of parallel links is connected to form a second of the pair of nested hinged joints.
 - 19.** The targeting guide of claim 18, wherein one of the links in the second pair forming the outer joint is configured so that its length between the nested hinged joint, it forms, and the second hinge point can be adjusted to change the first distance.
 - 20.** The targeting guide of claim 16, wherein the first sleeve is configured to receive a surgical wire and the second sleeve is configured to receive and guide a drill bit.
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